

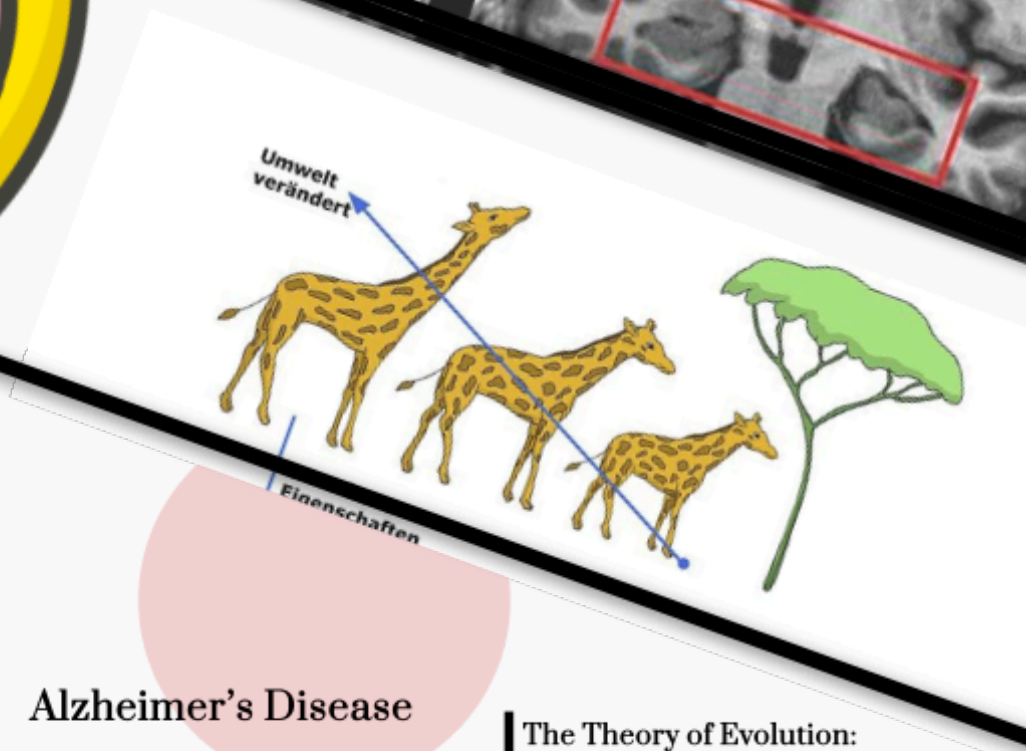
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STEM Action Teen Institution



Humor
Richard Wang

Alzheimer's Disease
Audrey Wu

**The Theory of Evolution:
Understanding Life's Diversity**
Kenny Wu

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The Development of Blue LEDs

By Yifan Yang

Our modern world is displayed through our many innovations, with our bright nights filled with lights and animated billboards. And none of it would've been possible, without the contributions of Shuji Nakamura, a man who dedicated years of his life, every single day without holidays or vacations, on perfecting the blue LED.

In the later 1900s, the red and green LED's had already been developed, being used in basic indicators and light ups. Unfortunately, lacking the blue in RGB prevents access to different color patterns that would fully allow us to create multicolored displays, leading to the very device you're using right now. Shuji Nakamura began as a researcher at Nichia in Japan, in the semiconductor department, focused on creating the inner circuits that LEDs use to provide light.

Semiconductors function using the atoms of different elements, each of the electrons occupying a different orbit, or energy band. A valence band and a conduction band create a jump for electrons to switch between levels, the empty spaces left behind allow positive charges to be carried through. Falling back from the conduction band to the valence band, releases the electrons' spare energy that allows them to stay in the conduction band. This energy is what we see as visible light, and the bigger the jump was, the more energy was released, allowing for the emission of photons of different frequencies, creating various colors with different materials.

Nakamura would then commit to a venture, traveling to Gainesville to learn how to operate and create his own Metal-Organic Chemical Vapor Deposition, something he will use to create the crystal layers and lattices needed for semiconductors. Making his own adjustments and innovating existing methods, he was able to utilize Gallium Nitride to create the materials he needed to make an efficient and profitable crystal layer, allowing for the creation of the blue LED.

Humor

By Richard Wang

Remember that corny joke your teacher may have said that cracked a slight smile or maybe even a laugh on how corny it sounded. Humans have laughed for many centuries. In day to day conversations, there is likely laughter in it. Most of the time, this laughter is incited by a person's humor, the capacity to express and discern something that is funny. It acts as help in stressful or awkward situations. Humor is a very broad word though as there are a variety of types of humor that people have. This stimulus that creates laughter, comes with social and mental benefits as well.

Something somebody says may not be funny to you because people have different types of humor. Some common ones include self-deprecating humor. This involves the speaker putting themselves down to make others feel good and laugh. Another type of humor is deadpan humor where the speaker makes an absurd remark with a blunt and neutral reaction. Dark humor, this includes taboo topics which can be offensive to others but can also provoke a laugh. Satire another type of humor uses sarcasm and ironic

comments to explain their point in a more interesting way. Wordplay humor, the humor that makes people groan at how corny it was. This includes dad jokes and puns that use



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words and their meanings in clever ways. The listed types of humors aren't the only types of humors as there are many others. However, the ones listed are common types of humor that is seen in our modern day.

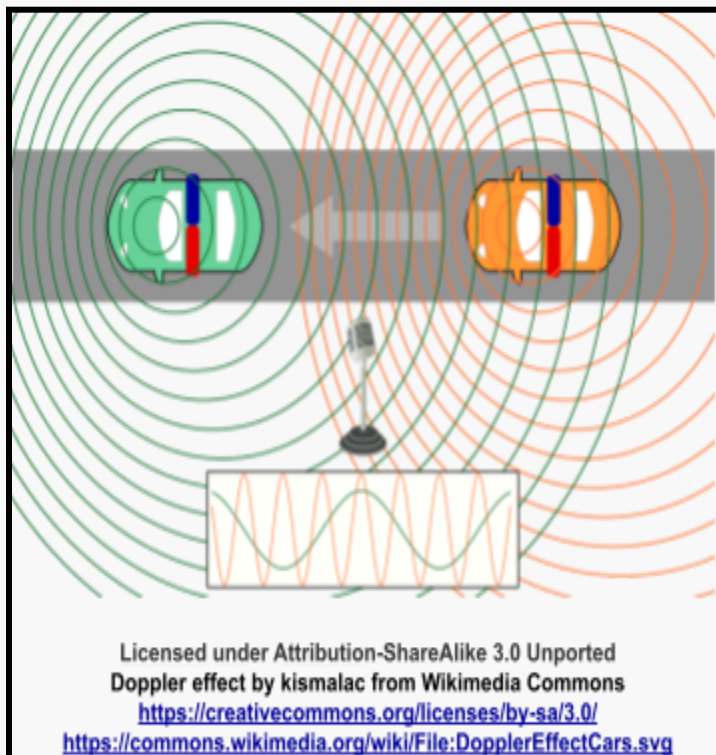
These types of humors are encountered in human communication but there are many theories on why do people find these types of humors funny. Humor, this simple 5 letter word we constantly experience, yet it is a complex aspect of human life that is hard to explain why it evokes the responses it does. There is a solid theory though that we laugh because humor works by throwing your expectations. The unexpected remark we hear creates this positive response of laughter.

At the end of the day, humor is essential in creating and consolidating relationships. It acts as a coping tool that remedies stress. With this in mind, people do not always receive humorous comments like others. Some may feel offense, so one's humor can act as a double edged blade when it comes to relationships. As great as humor is to social life, other people's sensitivities to comments should be considered.

The Doppler Effect and How It Shapes What We Hear and See

By Aidan Hong

The Doppler Effect is a fascinating scientific phenomenon that explains why the sound of a siren or a speeding car changes pitch as it passes by. First described by Austrian physicist Christian Doppler in 1842, the effect occurs when a wave source moves relative to an observer. Although it is most commonly associated with sound, the Doppler Effect also applies to light and other types of waves, making it a valuable concept in many fields of science and technology.



When a source of sound moves toward a listener, the sound waves are compressed, resulting in a higher frequency or pitch. As the source moves away, the waves are stretched, leading to a lower pitch. This is why an ambulance siren sounds high as it approaches and suddenly drops in pitch as it moves away. The actual sound produced by the siren doesn't change—only the way it is perceived by someone standing still. This change in frequency due to motion is the essence of the Doppler Effect.

The Doppler Effect is not limited to sound; it also affects light. Astronomers use this principle to determine whether stars or galaxies are moving toward or away from Earth. When an object in space moves away from us, its light is shifted to longer wavelengths—a phenomenon known as redshift. Conversely, if it is moving closer, the light shifts to shorter wavelengths, called blueshift. These observations have been key in supporting the theory that the universe is expanding.

In addition to astronomy, the Doppler Effect has many practical applications. Police use radar guns based on this principle to measure the speed of vehicles. Meteorologists rely on Doppler radar to track weather patterns, such as the movement of rain or storms. In medicine, Doppler ultrasound is used to observe blood flow and detect blockages in arteries and veins.

In conclusion, the Doppler Effect is a simple yet powerful concept that helps explain how waves behave when there is movement involved. Whether it's the pitch of a siren, the color of a distant star, or the speed of a moving car, the Doppler Effect plays a key role in helping us understand and measure the world around us.

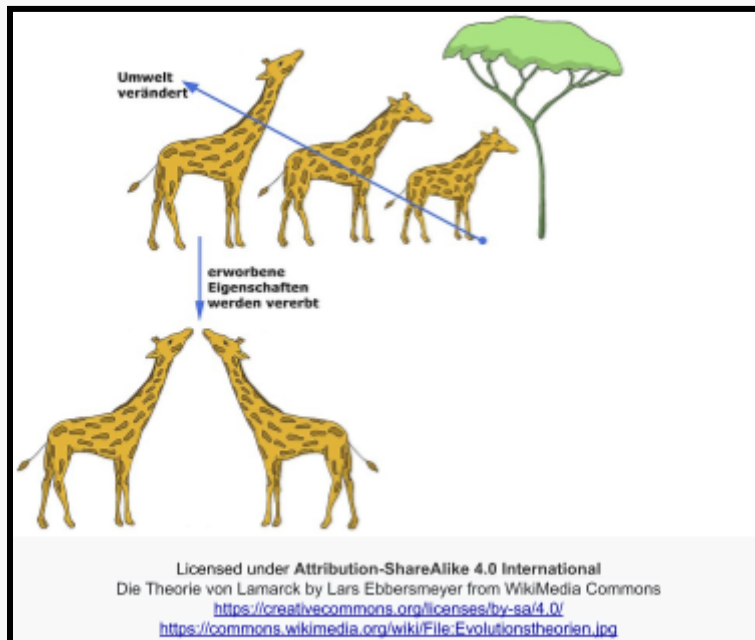
The Theory of Evolution: Understanding Life's Diversity

By Kenny Wu

The theory of evolution is one of the most important and widely accepted scientific explanations for the diversity of life on Earth. Proposed by Charles Darwin in the 19th century, the theory suggests that all species of living organisms have descended over time from common ancestors. Through a process called natural selection, traits that help organisms survive and reproduce become more common in a population, leading to

gradual changes and the development of new species.

At the heart of the theory is the idea that individuals within a species vary in their traits. Some of these traits are better suited to the environment, giving those individuals a better chance of surviving and passing on their genes. Over many generations, these advantageous traits become more common. For



example, giraffes with slightly longer necks were more likely to reach food high in trees and survive, leading to the long-necked giraffes we see today.

Evidence for evolution comes from many areas of science. Fossil records show how species have changed over millions of years. Comparative anatomy reveals

similarities between different organisms, such as the bones in a human hand, a bird wing, and a whale flipper—all of which point to a common ancestor. Genetics has further confirmed evolutionary theory by showing that all living things share a universal genetic code, and closely related species have more DNA in common.

While the theory of evolution was once controversial, especially due to religious and philosophical beliefs, it is now a central concept in biology. It helps scientists understand not only how species change over time but also how to study diseases, develop new medicines, and protect endangered species. Evolution also explains why living organisms are so well adapted to their environments.

In conclusion, the theory of evolution offers a powerful explanation for the origin and development of life on Earth. Through natural selection and adaptation, species have evolved over time in response to their surroundings. As more evidence continues to support Darwin's theory, evolution remains a cornerstone of modern science and an essential part of understanding the natural world.

Black Holes: The Mysterious Giants of Space

By Owen Chen

Black holes are among the most fascinating and mysterious objects in the universe. They are regions in space where gravity is so strong that nothing—not even light—can escape. Formed from the collapsed cores of massive stars, black holes challenge our understanding of physics, time, and space. Though invisible to the naked eye, scientists have found ways to detect and study these cosmic giants.

When a star much larger than our Sun runs out of fuel, it can no longer support itself against gravity and collapses inward. If the star is massive enough, this collapse continues until all its mass is concentrated into an extremely small point called a singularity. Around this singularity is the event horizon—the boundary beyond which nothing can return. Anything that crosses this point is lost to the black hole forever.

Even though we cannot see black holes directly, astronomers detect them by observing their effects on nearby objects. For example, when a black hole pulls in gas and dust from a nearby star, the material heats up and glows brightly, giving off X-rays that telescopes can detect. In 2019, scientists made history by capturing the first-ever image of a black hole's shadow using a global network of telescopes called the Event Horizon Telescope.

Black holes also teach us about the limits of our current understanding of physics. According to Albert Einstein's theory of general relativity, black holes warp space and time in extreme ways. Some scientists even believe that studying black holes could unlock secrets about the beginning of the universe, or possibly even lead to discoveries about time travel or parallel worlds—though these ideas remain purely theoretical.

In conclusion, black holes are powerful and mysterious forces in the universe. They mark the end of life for massive stars but open the door to new questions about the nature of space, time, and reality itself. As technology improves, our ability to study black holes will grow, helping us uncover more about these invisible giants and the universe we live in.



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Alzheimer's Disease

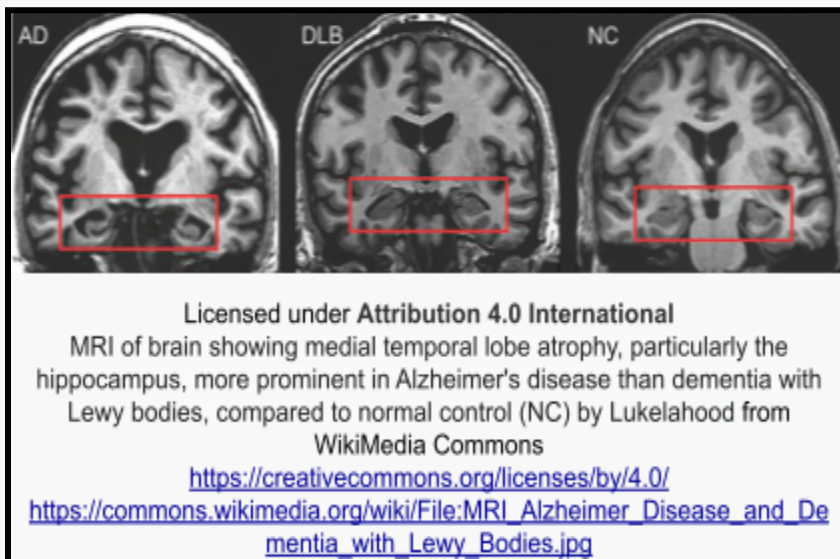
By Audrey Wu

Alzheimer's disease is a progressive neurological disorder that affects memory, thinking, and behavior. It is the most common cause of dementia, a general term for a decline in cognitive abilities severe enough to interfere with daily life. Alzheimer's typically begins with mild memory loss and confusion but advances over time to severe cognitive impairment and loss of independence. The disease most often affects older

adults, although

early-onset forms can occur in younger individuals.

At the biological level, Alzheimer's is associated with the buildup of two abnormal protein structures in the brain: amyloid plaques and tau tangles. Amyloid plaques



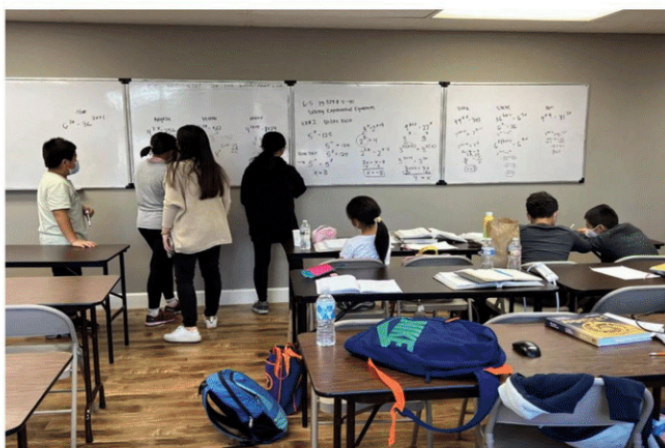
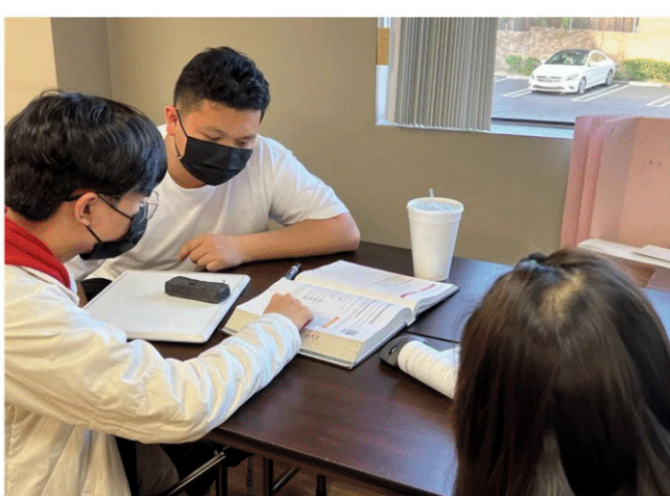
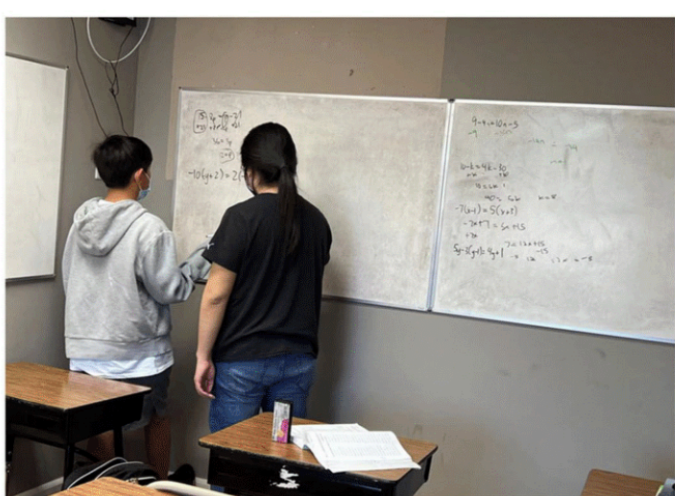
form between nerve cells, while tau tangles develop inside them. These abnormalities disrupt communication between neurons and eventually lead to cell death. As brain cells die, affected regions shrink—a process known as brain atrophy. This degeneration particularly impacts the hippocampus, which is crucial for forming new memories.

The exact causes of Alzheimer's are not fully understood, but a combination of genetic, environmental, and lifestyle factors likely contribute. The presence of the APOE-e4 gene variant increases the risk, although having it does not guarantee that a person will develop the disease. Other risk factors include age, family history,

cardiovascular disease, and head injuries. Some research suggests that healthy habits such as regular exercise, a balanced diet, and mental stimulation may help lower the risk or delay the onset.

There is currently no cure for Alzheimer's, but treatments are available to help manage symptoms. Medications such as cholinesterase inhibitors and memantine can improve communication between nerve cells and temporarily slow symptom progression. In addition, non-drug approaches—like creating structured routines, providing cognitive therapies, and ensuring a safe environment—can significantly improve the quality of life for those affected. Support for caregivers is also a vital component of care, as the disease places considerable emotional and physical demands on them.

Research into Alzheimer's is ongoing, with scientists exploring a wide range of potential therapies aimed at altering disease progression rather than just treating symptoms. Recent advances in imaging and biomarkers have made early detection more feasible, and clinical trials are testing drugs that target amyloid and tau directly. While the path to a cure remains challenging, growing global awareness and investment in research offer hope for future breakthroughs in understanding and combating Alzheimer's disease.



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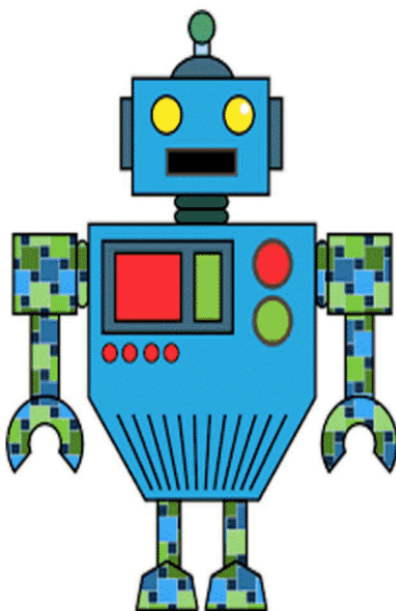
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